Step 10 One-Pager — Population-Based Training and Evolutionary Game Theory

Research on the possibilities for applying Artificial Intelligence in computer games

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Problem. Self-play can go in circles: in a cyclic game (rock-paper-scissors) "get better" has no meaning, so a population that only trains against itself spins instead of improving. Step 10 is the population-level layer of the thesis — how to *diagnose* whether a game/population is a skill ladder or a wheel of counters, and how to *train and evaluate* a whole population of agents. It is the launch point for automated opponent modeling (#1), population safe-exploitation without a minimax anchor (#2), and population-level evaluation (#3).

Approach. Two halves, both graded against exact references. **Part I — evolutionary game theory as a diagnostic:** replicator dynamics on four solvable matrix games (checked against analytic ESS/Nash) and the transitive/cyclic **spinning-top** decomposition. **Part II — an AlphaStar-style PBT league** of neural PPO agents on Leduc Hold'em (three agent types: main / main-exploiter / league-exploiter, plus freezing and PFSP), evaluated with **EGTA/meta-Nash**, Elo, and diversity metrics — every neural policy extracted to a *tabular* policy so Step 07's **exact** best response measures its exploitability. **All numbers below are measured.**

Key results (measured).

- *Replicator dynamics reproduce every analytic outcome.* Prisoner's Dilemma → all-Defect; Hawk-Dove → the interior 0.5 ESS (orbit radius 0.0); **Rock-Paper-Scissors never converges** (orbits the centre, radius 0.095); Stag Hunt → a basin-dependent pure ESS.
- *The spinning-top diagnostic works, and Leduc's structure depends on the population.* Hodge decomposition gives RPS transitive 0.0 / cyclic 1.0 and a skill ladder 1.0 / 0.0 (the SVD rank-1 method wrongly gives RPS 0.707). The **PSRO best-response** meta-game on Leduc is mostly **cyclic** ($\approx$ 0.45 transitive, 27 three-cycles); the **league snapshot** meta-game is mostly **transitive** ($\approx$ 0.94-0.98) — a contradicted prediction, §4/§7 of the report.
- *The league produces strong individuals.* Its best frozen snapshot reaches exploitability 1.305 (scale), beating exact PSRO (2.163) and self-play (3.683); CFR-Nash floor is 0.0099.
- *Honest negatives (kept predictions).* League exploitability is **non-monotone at scale** (4.73 $\rightarrow$ min $\approx$ 1.21 $\rightarrow$ 2.05 over 120 epochs) — the live agents regress late; the best agents are frozen snapshots. And the **meta-Nash mixture is *more* exploitable than its best member** at scale (3.418 > 1.305) — meta-Nash minimizes meta-game regret, not full-game exploitability, so mixing behavioral policies can hurt. Diversity is thin (participation ratio 1.9, a single behavioral cluster).

Thesis connection. The league is Step 9's PSRO made asynchronous with neural oracles, reusing Step 07's exact best response as the yardstick; main exploiters are automated opponent modelers (Contribution #1); EGTA/meta-Nash is the population evaluation methodology (Contribution #3). The league's missing guarantee — it can regress, and its mixture can be exploitable — is the population form of the **missing $N > 2$ safety anchor** (Contribution #2). The transitive/cyclic diagnostic predicts Step 11's FFA coalition games will be strongly cyclic.

Open questions. Does best-snapshot retention / population regularization remove the late regression (a training fix) or is it inherent to the three-type design (a design fix)? Should a population ship a selected best-response-robust member or a diversity-regularized meta-solver instead of the meta-Nash mixture? And underneath both: with no minimax value for a population, what does "safe" mean, and can the exploiter mechanism be given a guarantee rather than staying heuristic (Contribution #2)?